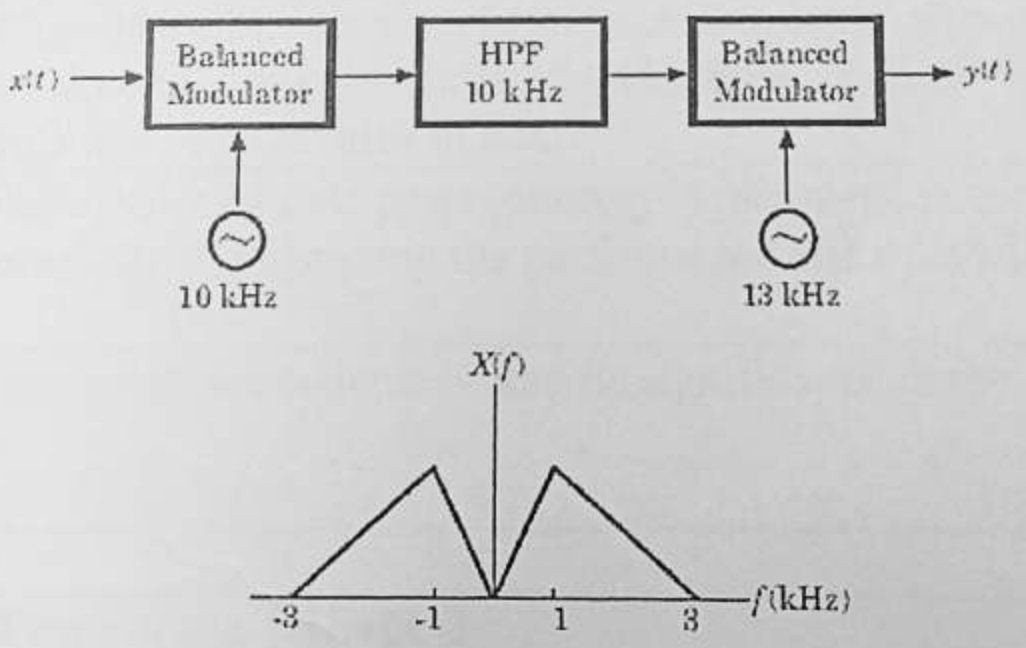
(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(Course Name : B. Tech. ECE/Reappear) (Semester : 4)
(March, 2024) OFF LINE mode

Subject Code: BEC-208	Subject: Communication System
Time : 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Given that, $f_X(x)=ae^{-b\|x\|}$, where x is a continuous random variable. Find the relation between a and b.	UNIT-1	
	(b) Describe the Gaussian Distribution and include a diagram to illustrate its concept.	UNIT-1	
	(c) Name and explain the two primary communication resources. Also, explain the objective pertaining to those.	UNIT-2	
	(d) For a signal amplitude modulated to a depth of 100% by a sinusoidal signal, how much is the power required for the modulated signal in terms of the power of the unmodulated carrier?	UNIT-2	
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a) A random process is given by X(t) = A cos(□o t +Θ), where A and □o are constants and Θ is a random variable, which is uniformly distributed in the interval (0, 2π). Find whether the given Random Process is Wide Sense Stationary or not?		
	(b) For a given function $x(t) = A\cos w_0 t$, find ACF, PSD and power.		
	(c) What are ergodic random processes? Explain ensemble averages and time averages in detail.		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) Consider a system as shown in figure. Let X(f) and Y(f) denote the fourier transforms of x(t) and y(t) respectively. The ideal HPF has cutoff frequency 10KHz. What are the positive frequencies where Y(f) has spectral peaks? 		
	(b) Discuss the concept of under-modulation, critical-modulation, and over-modulation, including their respective waveforms in the time domain.		
	(c) Explain the working of the Square law modulator along with its block diagram.		